Solutions to the additional problems

1. (a) For the Laplace distribution we have the probability density function

$$f_X(x) = \frac{1}{2}e^{-|x|}.$$

By definition the moment generating function will be given as

$$M_X(\theta) = \mathbb{E}[e^{\theta X}].$$

For the Laplace distribution this is

$$M_X(\theta) = \frac{1}{2}\int_{-\infty}^{\infty} e^{\theta x}e^{-|x|}dx$$

[1 mark]

We split this into two cases:

$$M_X(\theta) = \frac{1}{2}\int_{-\infty}^{0} e^{\theta x}e^{x}dx + \frac{1}{2}\int_{0}^{\infty} e^{\theta x}e^{-x}dx = \frac{1}{2}\int_{-\infty}^{0} e^{(\theta+1)x}dx + \frac{1}{2}\int_{0}^{\infty} e^{(\theta-1)x}dx.$$

[1 mark]

Note both integrals are finite if and only if $-1 < \theta < 1$. Indeed, $x \mapsto e^{\lambda x}$ is integrable over $[0, \infty)$ if and only if $\lambda < 0$. Therefore $e^{(\theta-1)x}$ is integrable over $[0, \infty)$ if only if $\theta < 1$ and $e^{(\theta+1)x}$ is integrable over $[0, \infty)$ if only if $\theta > -1$. Thus the only values of θ for which both integrals are finite is $-1 < \theta < 1$.

[1 mark for correctly stating values of θ]

For $\theta \in (-1, 1)$ we have

$$M_X(\theta) = \frac{1}{2}\left[\frac{1}{\theta+1}e^{(\theta+1)x}\right]_{x=-\infty}^{x=0} + \frac{1}{2}\left[\frac{1}{\theta-1}e^{(\theta-1)x}\right]_{x=0}^{x=+\infty}.$$

[1 mark]

Evaluating the limits

$$M_X(\theta) = \frac{1}{2}\left(\frac{1}{\theta+1} - \frac{1}{\theta-1}\right).$$

[1 mark]

Rearranging gives

$$M_X(\theta) = \frac{1}{1-\theta^2}.$$

[1 mark]

(b) To determine the second moment we can use the geometric series to write

$$M_X(\theta) = \frac{1}{1-\theta^2} = 1 + \theta^2 + \theta^4 + \dots$$

[1 mark]

In general, the moment generating function has the power series

$$M_X(\theta) = \sum_{k=0}^{\infty} \frac{\mathbb{E}[X^k]}{k!}\theta^k.$$

[1 mark]

Therefore equating θ^2 terms we see

$$\mathbb{E}[X^2] = 2! = 2.$$

and by using symmetry or by equating order θ terms we see that

$$\mathbb{E}[X] = 0.$$

[1 mark]

Therefore the variance is

$$\mathbb{V}\text{ar}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = 2.$$

[1 mark].

[Alternatively using the first and second derivatives of the moment generating function to calculate the variance also gives the 4 marks]

[Total 10 marks]

2. (a) From the definition of the moment generating function

$$M_{X+Y} = \mathbb{E}\left[e^{\theta(X+Y)}\right].$$

Using properties of the exponential we can write this as

$$M_{X+Y} = \mathbb{E}\left[e^{\theta X}e^{\theta Y}\right].$$

Since X and Y are independent

$$M_{X+Y} = \mathbb{E}\left[e^{\theta X}\right]\mathbb{E}\left[e^{\theta Y}\right] = M_X(\theta)M_Y(\theta).$$

Using the given expression for the moment generating functions of X and Y we have for $\theta < 1/3$

$$M_{X+Y} = \frac{1}{1-3\theta}\frac{1}{(1-3\theta)^2} = \frac{1}{(1-3\theta)^3}.$$

(b) We can calculate moments of a random variable using

$$M_Y^{(k)}(0) = \mathbb{E}[Y^k].$$

Calculating the derivatives:

$$\begin{aligned} M_Y(\theta) &= (1 - 258\theta)^{-3} \\ M_Y'(\theta) &= (-258)(-3)(1 - 258\theta)^{-4} \\ M_Y''(\theta) &= (-258)^2(-3)(-4)(1 - 258\theta)^{-5}. \end{aligned}$$

Now setting $\theta = 0$:

$$\begin{aligned} M_Y(0) &= 1 \\ \mathbb{E}[Y] = M_Y'(0) &= (-258)(-3) = 774 \\ \mathbb{E}[Y^2] = M_Y''(0) &= (-258)^2(-3)(-4) = 798768. \end{aligned}$$

Therefore the mean is $\mathbb{E}[Y] = 774$ and the variance is

$$\mathbb{V}\text{ar}(Y) = \mathbb{E}[Y^2] - \mathbb{E}[Y]^2 = 798768 - 774^2 = 199692.$$

[5 marks]

[Total 9 marks]

3. (a) We start by calculating the expected value:

$$\begin{aligned} \mathbb{E}[X] &= \int_{-\infty}^{\infty} x f_X(x) dx \\ &= \int_0^1 x \left(\lambda \left(x - \frac{1}{2} \right) + 1 \right) dx \\ &= \int_0^1 \left(\lambda x^2 - \frac{\lambda x}{2} + x \right) dx \\ &= \left[\frac{\lambda x^3}{3} - \frac{\lambda x^2}{4} + \frac{x^2}{2} \right] \\ &= \frac{\lambda}{3} - \frac{\lambda}{4} + \frac{1}{2} \\ &= \frac{\lambda + 6}{12}. \end{aligned}$$

[3 marks]

Now the bias of $\hat{\lambda}$ is given by

$$\begin{aligned} \text{Bias}(\hat{\lambda}) &= \mathbb{E}[\hat{\lambda}(\underline{X})] - \lambda \\ &= \mathbb{E}[12\bar{X} - 6] - \lambda \\ &= 12\mathbb{E}[\bar{X}] - 6 - \lambda. \end{aligned}$$

Since $X_1, \ldots, X_n$ are identically distributed we have

$$\mathbb{E}[\bar{X}] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] = \mathbb{E}[X] = \frac{\lambda+6}{12}.$$

Therefore the bias of $\hat{\lambda}$ is

$$\text{Bias}(\hat{\lambda}) = 12\left(\frac{\lambda+6}{12}\right) - 6 - \lambda = 0.$$

Hence $\hat{\lambda}$ is an unbiased estimator of λ.

[3 marks]

(b) We calculate the second moment as

$$\begin{aligned}
\mathbb{E}[X^2] &= \int_{-\infty}^{\infty} x^2 f_X(x)dx \\
&= \int_0^1 x^2\left(\lambda\left(x-\frac{1}{2}\right)+1\right)dx \\
&= \int_0^1 \left(\lambda x^3 - \lambda\frac{x^2}{2} + x^2\right)dx \\
&= \left[\frac{\lambda x^4}{4} - \frac{\lambda x^3}{6} + \frac{x^3}{3}\right]_{x=0}^{1} \\
&= \frac{\lambda}{4} - \frac{\lambda}{6} + \frac{1}{3} = \frac{\lambda+4}{12}.
\end{aligned}$$

[2 marks]

Using this and the calculation in part (a) we have

$$\begin{aligned}
\mathbb{V}\text{ar}(X) &= \mathbb{E}[X^2] - \mathbb{E}[X]^2 \\
&= \frac{\lambda+4}{12} - \frac{(\lambda+6)^2}{144} = \frac{12-\lambda^2}{144}
\end{aligned}$$

[2 marks]

Now we can find the MSE as

$$\begin{aligned}
\text{MSE}(\hat{\lambda}) &= \mathbb{V}\text{ar}(\hat{\lambda}) + \text{Bias}(\hat{\lambda})^2 \\
&= \mathbb{V}\text{ar}(\hat{\lambda}) \\
&= \mathbb{V}\text{ar}(12\bar{X} - 6) \\
&= 12^2\mathbb{V}\text{ar}(\bar{X}) \\
&= 144\mathbb{V}\text{ar}\left(\frac{1}{n}\sum_{i=1}^{n}X_i\right) \\
&= \frac{144}{n^2}\sum_{i=1}^{n}\mathbb{V}\text{ar}\left(X_i\right) \qquad \text{using independence} \\
&= \frac{144}{n}\frac{12-\lambda^2}{144} = \frac{12-\lambda^2}{n}
\end{aligned}$$

[3 marks]

(c) $\hat{\lambda}$ is a good estimator because it is unbiased and the MSE($\hat{\lambda}$) goes to zero as n tends to infinity.

[1 mark]

[Total 14 marks]

4. (a) The bias of an estimator is given:

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$$

$$\begin{aligned}
\text{Bias}\left[\hat{\theta}_1\right] &= \mathbb{E}\left[\frac{X_1 + X_2 + X_3}{3}\right] - \theta \\
&= \frac{\mathbb{E}[X_1] + \mathbb{E}[X_2] + \mathbb{E}[X_3]}{3} - \theta \\
&= \frac{\theta + \theta + \theta}{3} - \theta \\
&= 0
\end{aligned}$$

[2 marks]

$$\begin{aligned}
\text{Bias}\left[\hat{\theta}_2\right] &= \mathbb{E}\left[\frac{X_1 + 2X_2 + 3X_3}{3}\right] - \theta \\
&= \frac{\mathbb{E}[X_1] + 2\mathbb{E}[X_2] + 3\mathbb{E}[X_3]}{3} - \theta \\
&= \frac{\theta + 2\theta + 3\theta}{3} - \theta \\
&= \frac{6\theta}{3} - \theta \\
&= 2\theta - \theta \\
&= \theta
\end{aligned}$$

[2 marks]

Therefore, we can conclude that θ_1 is an unbiased estimator while θ_2 is a biased estimator of θ.

[1 mark]

(b) The Mean Squared Error (MSE) of an estimator is given by:

$$\text{MSE}(\hat{\theta}) = \mathbb{V}\text{ar}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

$$\begin{aligned}
\mathbb{V}\text{ar}\left(\hat{\theta}_2\right) &= \mathbb{V}\text{ar}\left(\frac{X_1 + 2X_2 + 3X_3}{3}\right) \\
&= \frac{\mathbb{V}\text{ar}\left(X_1\right) + 2^2\mathbb{V}\text{ar}\left(X_2\right) + 3^2\mathbb{V}\text{ar}\left(X_3\right)}{3^2} \\
&= \frac{\theta + 2^2\theta + 3^2\theta}{9} \\
&= \frac{\theta(1+4+9)}{9} \\
&= \frac{14\theta}{9}
\end{aligned}$$

[2 marks]

$$\begin{aligned}
\text{MSE}\left(\hat{\theta}_2\right) &= \mathbb{V}\text{ar}\left(\hat{\theta}_2\right) + \left[\text{Bias}\left(\hat{\theta}_2\right)\right]^2 \\
&= \frac{14\theta}{9} + \theta^2 \\
&= \frac{\theta(14+\theta)}{9}
\end{aligned}$$

[1 mark]

(c) The efficiency of θ_1 relative to θ_2 is given by:

$$\begin{aligned}
\text{Efficiency} &= \frac{\text{MSE}\left(\hat{\theta}_1\right)}{\text{MSE}\left(\hat{\theta}_2\right)} \\
&= \frac{\frac{\theta}{3}}{\frac{\theta(14+\theta)}{9}} \\
&= \frac{3}{14+\theta}
\end{aligned}$$

[1 mark]

Since the efficiency is less than 1 for all values of θ, we can conclude that θ_1 is more efficient compare to θ_2.

[1 mark]

[Total 10 marks]

5. (a) The maximum likelihood estimator (MLE) for λ:
The likelihood function is given by:

$$L(\lambda) = \prod_{i=1}^{n} f(x_i) = \prod_{i=1}^{n} \lambda x_i^{\lambda-1}$$
$$= \lambda^n \left(\prod_{i=1}^{n} x_i \right)^{\lambda-1}$$

[1 mark]

Taking the natural logarithm to get the log-likelihood:

$$\ell(\lambda) = \log L(\lambda) = n \log \lambda + (\lambda - 1) \sum_{i=1}^{n} \log x_i$$

[1 mark]

Differentiate $\ell(\lambda)$ with respect to λ and set it equal to zero to find the maximum likelihood estimator (MLE) for λ:

$$\frac{d\ell(\lambda)}{d\lambda} = \frac{n}{\lambda} + \sum_{i=1}^{n} \log x_i$$
$$\frac{n}{\lambda} + \sum_{i=1}^{n} \log x_i = 0$$

[1 mark]

Solving for λ :

$$\frac{n}{\lambda} = -\sum_{i=1}^{n} \log x_i$$
$$\hat{\lambda}_{\text{MLE}} = -\frac{n}{\sum_{i=1}^{n} \log x_i}$$

[1 mark]

It remains to verify that this is indeed the maximum, for this we use the second derivative test (or a suitable plot of the function):

$$\frac{d^2\ell}{d\lambda^2} = -\frac{n}{\lambda^2} < 0$$

Since this is always negative it must also be negative at the critical point, showing that the critical point is a local maximum.

[1 mark]

(b) The method of moments estimator (MME) for λ:
Equate the population moment to the sample moment:

$$E(X) = \mu_1$$

$$\begin{aligned}\mathbb{E}[X] = \int_0^1 x f(x) dx &= \int_0^1 x\lambda x^{\lambda-1} dx \\ &= \lambda \int_0^1 x^{\lambda} dx \\ &= \frac{\lambda}{\lambda+1}\end{aligned}$$

[2 marks]

$$\mathbb{E}(X) = \mu_1$$

$$\mathbb{E}(X) = \frac{1}{1+\lambda} \text{ and } \mu_1 = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$$

$$\frac{1}{1+\lambda} = \bar{x}$$

[1 mark]

Solving for λ :

$$\hat{\lambda}_{\text{MME}} = \frac{1-\bar{x}}{\bar{x}}$$

[1 mark]

(c) Given the data values we have:

$$\hat{\lambda}_{\text{MLE}} = -\frac{n}{\sum_{i=1}^{n} \log(x_i)} = -\frac{3}{\log(0.1) + \log(0.2) + \log(0.3)} \approx 0.586$$

[1 mark]

$$\bar{x} = \frac{0.1 + 0.2 + 0.3}{3} \approx 0.2$$

$$\hat{\lambda}_{\text{MM}} = \frac{1-\bar{x}}{\bar{x}} = \frac{1-0.2}{0.2} \approx 0.25$$

The MLE estimator provides the smallest estimate for λ.

[1 mark]

[Total 10 marks]

6. (a) The likelihood is given by

$$\begin{aligned} L(\beta; \underline{x}) &= \prod_{i=1}^{n} f(x_i; \beta) \\ &= \prod_{i=1}^{n} \beta^2 x_i e^{-\beta x_i} \\ &= \beta^{2n} e^{-\beta \sum_{i=1}^{n} x_i} \prod_{i=1}^{n} x_i. \end{aligned}$$

Now taking the logarithm gives the log-likelihood as

$$\ell(\beta; \underline{x}) = \log(L(\beta; \underline{x})) = 2n \log(\beta) + \sum_{i=1}^{n} \log(x_i) - \beta \sum_{i=1}^{n} x_i.$$

[3 marks]

(b) We can find the critical points of the log-likelihood by differentiating with respect to β and solving $\partial \ell / \partial \beta = 0$. Note that

$$\frac{\partial \ell(\beta; \underline{x})}{\partial \beta} = \frac{2n}{\beta} - \sum_{i=1}^{n} x_i.$$

At a critical point we have

$$\hat{\beta}(\underline{x}) = \frac{2n}{\sum_{i=1}^{n} x_i} = \frac{2}{\bar{x}}.$$

[2 marks]

To verify that this is a maximum of ℓ we note that

$$\frac{\partial^2 \ell(\beta; \underline{x})}{\partial \beta^2} = -\frac{2n}{\beta^2} < 0.$$

As this is negative for all β we have that $\hat{\beta}$ maximises ℓ. [1 mark]

(c) From the distribution formula sheet we have

$$\mathbb{E}[X] = \frac{2}{\beta}.$$

Which we can rearrange as

$$\beta = \frac{2}{\mathbb{E}[X]}.$$

The method of moments estimator is obtained by estimating $\mathbb{E}[X]$ with the sample mean $\bar{X}$ which gives

$$\hat{\beta}(\underline{X}) = \frac{2}{\bar{X}}.$$

For the given data we have

$$\bar{x} = \frac{0.13 + 0.21 + 0.18 + 0.33}{4} = 0.2125,$$
$$\hat{\beta}(\underline{x}) = \frac{2}{0.2125} = 9.412.$$

[4 marks]

[Total 10 marks]

7. (a) The quantity

$$\frac{\bar{X} - \mu}{S^2/n}$$

is a pivotal quantity for $\bar{X}$ because:

i. It is a function which involves the parameter μ and the sample.

[1 mark]

ii. It has a known distribution which does not depend on the parameter μ. That is, the quantity follows a t-distribution with $n-1$ degrees of freedom.

[1 mark]

iii. It is a monotonic function of the parameter μ.

[1 mark]

(b) The 95% confidence interval for μ is given by

$$\begin{aligned}
&\left(\bar{x} - t_{n-1,0.025}\sqrt{s^2/n}, \bar{x} + t_{n-1,0.025}\sqrt{s^2/n}\right) \\
&= \left(50.05 - t_{14,0.025}\sqrt{0.015/15}, 50.05 + t_{14,0.025}\sqrt{0.015/15}\right) \\
&= \left(50.05 - 2.145\sqrt{0.015/15}, 50.05 + 2.145\sqrt{0.015/15}\right) \\
&= (49.982, 50.118).
\end{aligned}$$

[2 marks]

The calculated 95% confidence interval for the mean diameter is (49.982, 50.118) mm, which includes the target value of 50 mm. The new process does not appear to lead to a significantly different mean diameter from the target of 50 mm based on this sample.

[1 mark]

(c) For a 95% confidence interval for σ^2 we need the critical values from the χ^2-distribution:

$$\chi^2_{n-1,1-\alpha/2} = \chi^2_{14,0.975} = 5.629$$
$$\chi^2_{n-1,\alpha/2} = \chi^2_{14,0.025} = 26.12$$

[1 mark]

The 95% confidence interval for σ^2 is given by:

$$\left(\frac{(n-1)s^2}{\chi^2_{14,0.025}}, \frac{(n-1)s^2}{\chi^2_{14,0.975}}\right) = \left(\frac{(14)(0.015)}{26.12}, \frac{(14)(0.015)}{5.629}\right) = (0.008, 0.037).$$

[2 marks]

Since the 95% confidence interval for the variance $(0.008, 0.037)\text{mm}^2$ includes the population variance value of 0.02mm^2, we cannot conclude that there is a significant difference in the variability of the component diameters produced by the new process when compared to the target variance of 0.02mm^2.

[1 mark]

[Total 10 marks]

8. We add a row with the differences to the table:

Patient ID	1	2	3	4	5	6	7	8	9	10	11
Meter 1 (x)	60	88	73	61	49	69	93	71	94	66	54
Meter 2 (y)	62	88	75	61	48	72	90	74	93	63	55
$d = x - y$	-2	0	-2	0	1	-3	3	-3	1	3	-1

We can summarise the difference data as

$$\begin{aligned} n &= 11, \\ \bar{d} &= \frac{1}{n}\sum_{i=1}^{n} d_i = -0.273 \\ s_d &= \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(d_i - \bar{d})^2} = 2.149. \end{aligned}$$

[3 marks]

In order to construct a confidence interval we will assume that the differences $D_1, \ldots, D_n$ in meter readings are normally distributed.

[1 mark]

We will use the pivotal quantity

$$h(\underline{D}, \mu_d) = \frac{\bar{D} - \mu_d}{S_d/\sqrt{n}}$$

Note that since we are assuming $\underline{D}$ are normally distributed we may assume that $h(\underline{D}, \mu_d) \sim t_{10}$. From the statistical tables we have

$$t_{0.025,10} = 2.228.$$

[1 mark]

A 95% confidence interval for μ_d, the mean difference in meter readings is given by:

$$\left(-0.273 - 2.228\frac{2.149}{\sqrt{11}}, -0.273 + 2.228\frac{2.149}{\sqrt{11}}\right) = (-1.72, 1.17).$$

[2 marks]

The interval $(-1.72, 1.17)$ contains zero, therefore, the physician has no evidence to show that the two glucose meters give significantly different readings at the 5% significance level.

[1 mark]

[Total 8 marks]